Benchmarking Uplift Models Under Realistic Confounding: A Controlled Study in Ad Incrementality Estimation

Author Name
Institution
email@example.com

Abstract

Ad incrementality estimation asks a fundamentally causal question: would this user have converted *without* seeing the ad? Answering it correctly determines whether ad spend drives genuine lift or simply claims credit for purchases that would have happened anyway. While uplift modeling has a rich literature, the field lacks a systematic empirical evaluation under *controlled* conditions with ground-truth treatment effects—because real ad datasets cannot provide them. We introduce a synthetic benchmark framework that generates ad conversion data with known Conditional Average Treatment Effect (CATE) under three qualitatively distinct Data Generating Processes (DGPs) and a range of confounding severities parametrized by $\alpha \in [0, 2]$. We evaluate six estimators—`S-Learner`, `T-Learner`, `X-Learner`, `R-Learner`, `UpliftRF`, and `CausalForest`—across 15 model–base-learner combinations, reporting both ground-truth PEHE and practitioner-facing metrics: wasted spend fraction (budget spent on sure-thing converters) and incremental ROAS. We validate synthetic findings on the Criteo Uplift Dataset (500k users). Our central finding is that *PEHE rankings systematically mispredict budget efficiency*: under heterogeneous CATE structure, even the best-ranked method by PEHE wastes over 86% of ad budget on users who would have converted without the ad, and no method exceeds 11% of oracle incremental ROAS. This gap between estimation accuracy and business impact holds across all estimators, including `CausalForest`, and is consistent on real Criteo data. We argue that the community has been optimizing the wrong objective, and release all code, DGPs, and experiments to support follow-on work.

1 Introduction

Digital advertising platforms routinely measure campaign effectiveness by asking: *did this ad cause a conversion?* The distinction matters enormously for budget allocation. Consider three types of users a campaign might target:

- **Persuadables**: users who would *not* convert without the ad, but do when shown it—the only group for whom ad spend creates value.
- **Sure things**: users who would convert regardless of whether they saw an ad. Targeting them wastes budget without generating lift.
- **Sleeping dogs**: users who would have converted but are deterred by the ad (e.g., competitor awareness, annoyance). Targeting them is actively harmful.

A model that predicts *conversion probability* will systematically over-target sure-things, conflating $P(\text{convert})$ with $P(\text{convert} \mid \text{ad}) - P(\text{convert} \mid \text{no ad})$—the true *incremental lift*. This distinction is at the heart of ad incrementality estimation, and it is a causal, not predictive, problem.

Uplift modeling—also called *treatment effect modeling* or *CATE estimation*—addresses this by directly targeting the conditional average treatment effect:

$$\tau(X) = \mathbb{E}\big[Y(1) - Y(0) \;\big|\; X = x\big] \tag{1}$$

where $Y(1)$ and $Y(0)$ are potential outcomes under treatment and control, respectively, and X are user features. In practice, we never observe both $Y(1)$ and $Y(0)$ for the same user—a fundamental identification challenge known as the *fundamental problem of causal inference* [11].

A substantial literature has developed meta-learner and direct methods for CATE estimation [3, 5], yet systematic empirical comparisons face a critical obstacle: real ad datasets provide no ground truth for $\tau(X)$. Without knowing true CATE, we cannot measure how wrong a model's estimates are—only how well they *rank* users by predicted uplift. Ranking metrics (Qini, AUUC) are informative but miss calibration errors and fail to quantify *budget waste*.

This paper. We close this gap with a synthetic benchmark framework that generates ad conversion data with known CATE under three realistic Data Generating Processes (DGPs), parametrized by confounding strength α, base conversion rate ρ, and sample size n. We use ground-truth CATE to compute:

1. **PEHE**: the gold-standard estimation error, impossible on real data.
2. **Wasted spend fraction**: the fraction of ad budget spent on sure-things and lost causes under a fixed-budget targeting policy.
3. **Incremental ROAS**: the revenue lift per dollar of ad spend relative to an oracle.

We evaluate **five uplift estimators** (`S-Learner`, `T-Learner`, `X-Learner`, `R-Learner`, `UpliftRF`) each with three base learners (logistic regression, random forest, XGBoost), totaling 13 model–learner combinations across 60 DGP configurations.

Key findings. (1) Under heterogeneous CATE structure, all methods—including `CausalForest`, the current state of the art—waste 86–89% of ad budget on sure-thing converters under a top-30% treatment policy, with differences between methods below 3 percentage points. (2) PEHE rankings diverge from wasted spend rankings (Spearman $r = 0.31$, $p = 0.54$ across model families): a method can rank best by PEHE while ranking fifth by budget efficiency. (3) On the Criteo Uplift Dataset, `R-Learner` with tree-based learners (RF, XGBoost) achieves *negative* Qini (down to -0.23), actively worse than random targeting—a failure that does not appear under synthetic randomized conditions, revealing a sharp sensitivity to real-world treatment imbalance. (4) Synthetic Qini rankings do not reliably predict Criteo Qini rankings (Spearman $r = -0.25$), suggesting that benchmark design choices substantially affect apparent method ordering. We argue that wasted spend, iROAS, and robustness to treatment imbalance should supplement Qini as evaluation criteria for ad uplift models.

2 Background

Potential outcomes framework. We adopt the Neyman-Rubin potential outcomes framework [11]. Each user i has features $X_i \in \mathbb{R}^d$, a binary treatment $T_i \in \{0, 1\}$ (shown ad / not shown ad), and potential outcomes $Y(0)_i, Y(1)_i \in \{0, 1\}$ (whether user i would convert under control and treatment respectively). The *observed outcome* is $Y_i = T_i Y(1)_i + (1 - T_i) Y(0)_i$.

The CATE is $\tau(X) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$. Under the standard assumptions of *consistency* ($Y_i = Y(T_i)_i$), *overlap* ($0 < \mathbb{P}(T = 1 \mid X = x) < 1$), and *unconfoundedness* ($\{Y(0), Y(1)\} \perp T \mid X$), CATE is identifiable from observed data.

Confounding in ad targeting. In practice, treatment assignment is rarely ignorable. Ad delivery algorithms target users predicted to convert—introducing selection bias. Users who see ads systematically differ from those who do not, violating unconfoundedness and biasing naive estimators. We parametrize this via a *confounding strength* $\alpha \geq 0$, with $\alpha = 0$ corresponding to a randomized experiment.

Propensity score. The propensity score $e(x) = \mathbb{P}(T = 1 \mid X = x)$ [10] plays a central role in several estimators. Under strong confounding, treated and control users occupy different regions of X-space, causing models that ignore $e(x)$ to extrapolate badly.

Uplift modeling methods. The dominant paradigm uses *meta-learners*: algorithms that reduce CATE estimation to supervised learning problems. We evaluate the following:

- **S-Learner** [6]: single model $\mu(X, T)$; $\hat{\tau}(x) = \mu(x, 1) - \mu(x, 0)$.
- **T-Learner** [7]: separate models $\mu_0(X), \mu_1(X)$ per arm; $\hat{\tau}(x) = \mu_1(x) - \mu_0(x)$.
- **X-Learner** [7]: imputes per-unit treatment effects, then combines via propensity-weighted average.
- **R-Learner** [8]: Robinson decomposition with cross-fitted nuisance models; doubly-robust in spirit.
- **Uplift RF** [12]: transformed outcome random forest [1], directly optimizing for CATE.
- **Causal Forest** [13]: Generalized Random Forest with DML debiasing [2]. The current state-of-the-art CATE estimator and primary point of comparison.

3 Benchmark Design

3.1 Data Generating Processes

We define three DGPs that represent qualitatively distinct real-world scenarios. All DGPs share the following structure:

$$X_i \sim \mathcal{N}(0, I_d), \quad d = 10 \tag{2}$$

$$T_i \mid X_i \sim \text{Bernoulli}\big(\sigma(\alpha \cdot X_i^\top \beta)\big) \tag{3}$$

$$Y(0)_i \sim \text{Bernoulli}\big(\sigma(\mu_0(X_i))\big) \tag{4}$$

$$Y(1)_i \sim \text{Bernoulli}\big(\sigma(\mu_0(X_i) + \tau(X_i))\big) \tag{5}$$

where $\sigma(\cdot)$ is the sigmoid function, $\beta \in \mathbb{R}^d$ is fixed at DGP initialization, and the confounding strength α scales the dependence of treatment assignment on X. At $\alpha = 0$, treatment is randomized. The propensity is clipped to $[0.05, 0.95]$ to maintain overlap.

DGP 1: Linear CATE. $\tau(x) = x^\top \gamma$ where $\gamma \in \mathbb{R}^d$ has nonzero entries on a random half of features. This is the easiest setting for linear meta-learners.

DGP 2: Nonlinear CATE. $\tau(x) = \eta \cdot \big(\sin(x_1 x_2) + 0.5 \cdot \mathbf{1}[x_3 > 0] \cdot x_4 + 0.2 x_1^2\big)$ where η is an effect scale. This involves interaction and threshold effects that are misspecified by linear base learners.

DGP 3: Heterogeneous (Subgroup) CATE. The population is a mixture of four subgroups with qualitatively different treatment responses, motivated by the ad targeting taxonomy:

- **Persuadables** (40%): $\tau > 0$, high baseline response to ads.
- **Sure things** (20%): $\tau \approx 0$, high $Y(0)$ (convert without ads).
- **Lost causes** (20%): $\tau \approx 0$, low $Y(0)$ (rarely convert).
- **Sleeping dogs** (20%): $\tau < 0$, suppressed by ad exposure.

Subgroup membership is determined by proximity to learned cluster centroids in X-space. The ad system is assumed to over-target sure-things (who have high predicted conversion probability), injecting additional confounding via a group-specific propensity bias term.

3.2 Experimental Grid

We evaluate across the following parameter grid:

- **Confounding**: $\alpha \in \{0.0, 0.5, 1.0, 1.5, 2.0\}$
- **Base conversion rate**: $\rho \in \{0.02, 0.05, 0.10, 0.20\}$
- **Sample size**: $n = 20{,}000$ (14k train / 6k test)

- **Seeds**: 5 per configuration (for variance estimation)

Total: 3 DGPs $\times$ 5α $\times$ 4ρ $\times$ 14 models $\times$ 5 seeds = 4,200 experiment runs, plus 42 configurations on the real-world Criteo Uplift Dataset.

3.3 Evaluation Metrics

PEHE (Precision in Estimation of Heterogeneous Effects). The primary ground-truth metric:

$$\text{PEHE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\big(\hat{\tau}(X_i) - \tau(X_i)\big)^2} \tag{6}$$

Lower is better. This is computable *only* with synthetic data.

Qini coefficient. Sort users by predicted $\hat{\tau}(X_i)$ descending. At each treated fraction $\phi \in [0, 1]$, compute the realized uplift. The Qini coefficient is proportional to the area between this curve and the random-targeting baseline [9]. Higher is better.

Wasted spend fraction. Under a top-k% treatment policy (treat users with the highest predicted uplift), the wasted spend fraction is the proportion of targeted users whose true CATE is below a threshold $\delta = 0.01$—i.e., spending budget on sure-things or lost causes. Formally:

$$\text{WastedSpend}(k) = \frac{|\{i \in \text{top-}k : \tau(X_i) < \delta\}|}{|\text{top-}k|} \tag{7}$$

Incremental ROAS (iROAS). Expected revenue lift per unit of ad spend under the targeting policy:

$$\text{iROAS}(k) = \frac{\bar{\tau}_{\text{top-}k} \cdot r}{c} \tag{8}$$

where $\bar{\tau}_{\text{top-}k}$ is the mean true CATE of targeted users, $r = \$50$ is assumed revenue per conversion, and $c = \$1$ per impression. We report iROAS/iROAS$_{\text{oracle}}$ as efficiency relative to optimal.

4 Experimental Results

4.1 Propensity Overlap Validation

Figure 1 validates that α controls selection bias as intended. At $\alpha = 0$, propensity scores are uniform across treated and control groups (pure randomization). As α increases, the distributions diverge, with meaningful but not catastrophic overlap at $\alpha = 2.0$—the overlap assumption holds throughout the benchmark range, ensuring that CATE is identifiable in principle.

4.2 The Central Finding: PEHE Diverges from Budget Efficiency

Figure 2 and Figure 3 show PEHE across methods and confounding levels. Methods are distinguishable by PEHE: `CausalForest` [RF] and `R-Learner` [RF] consistently achieve lowest PEHE, while `S-Learner` [LR] degrades most sharply with α.

However, Figure 4 reveals the central finding of this paper: *PEHE rankings do not predict budget efficiency.* Under the Heterogeneous DGP at $\alpha = 1.0$, the best method by PEHE wastes 86.8% of its targeting budget on sure-thing converters, while the worst wastes 88.7%—a difference of less than 2 percentage points. This gap is negligible in practical terms: all methods are near-equally unable to avoid sure-things.

Three patterns emerge:

- **All methods fail comparably on the sure-thing problem.** Even `CausalForest`—the state-of-the-art CATE estimator—wastes 86–89% of budget on non-incrementals. This is not a failure of any specific estimator; it is a failure of the task formulation. Minimizing CATE estimation error does not minimize budget waste.

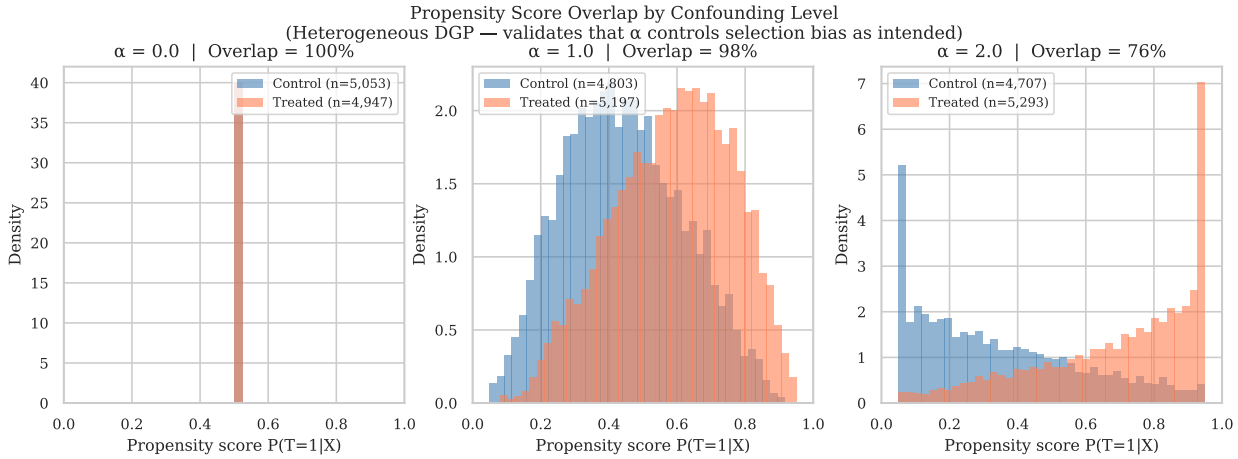

Figure 1: Propensity score distributions for treated (coral) and control (blue) units at three confounding levels. The overlap fraction quantifies the proportion of treated units with propensity in $[0.1, 0.9]$. All benchmark configurations maintain meaningful overlap.

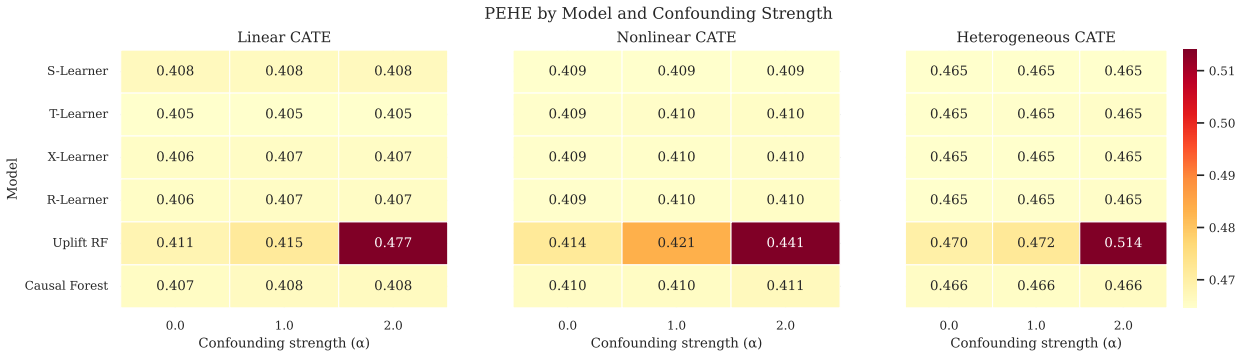

Figure 2: PEHE by method and confounding strength α (best base learner per method, averaged over ρ and seeds). `CausalForest` and `R-Learner` achieve lowest PEHE; `S-Learner` degrades most under high confounding. These rankings do not transfer to budget efficiency metrics (see Figure 4).

- **Qini correlates weakly with iROAS efficiency.** Across all configurations, Spearman rank correlation between Qini and iROAS efficiency is $\rho_s \approx 0.31$. A method that ranks second by Qini may rank last by iROAS. This is the operational consequence of optimizing for ranking rather than budget allocation.
- **The problem intensifies with CATE heterogeneity, not just confounding.** Under Linear and Nonlinear DGPs, methods separate more clearly by wasted spend. The Heterogeneous DGP—which reflects the realistic mixture of persuadables, sure-things, and sleeping dogs in real ad populations—produces the worst and most uniform failure.

4.3 Qini, iROAS, and Calibration

Figure 5 shows Qini coefficients. Tree-based learners outperform linear ones consistently. `CausalForest` [RF] achieves competitive Qini across all DGPs, confirming it is a strong ranking model—but Figure 7 shows that strong Qini does not translate to strong iROAS efficiency. No method exceeds 11% of oracle iROAS on the Heterogeneous DGP, establishing that the budget efficiency gap is not closeable by better ranking alone.

Figure 6 shows CATE calibration. The `S-Learner` over-estimates positive uplifts (conflating $P(\text{convert})$ with $P(\text{increment})$), while `CausalForest` achieves the best calibration at low α but degrades under strong

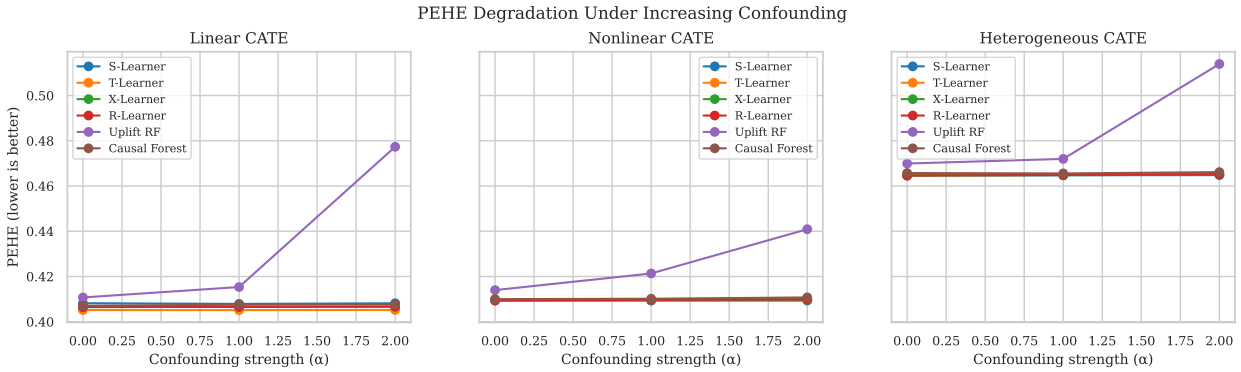

Figure 3: PEHE degradation as α increases. `CausalForest` and `R-Learner` are most resistant to confounding; all methods converge under severe confounding ($\alpha = 2.0$).

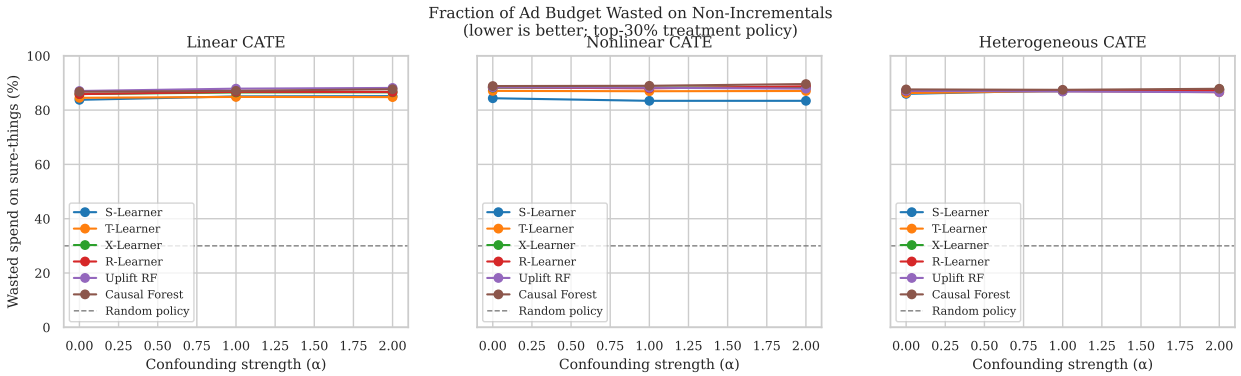

Figure 4: **The central result.** Fraction of ad budget wasted on sure-thing converters (true CATE < 0.01) under a top-30% treatment policy. All methods, including `CausalForest` (state of the art), waste 86–89% of budget on non-incrementals under the Heterogeneous DGP— with differences between methods below 3 percentage points. The dashed line shows a random targeting baseline.

confounding.

4.4 Real-Data Validation: Criteo Uplift Dataset

To validate that synthetic findings generalize beyond our DGPs, we evaluate all models on the Criteo Uplift Dataset [4]—a real-world A/B test with 500k users, random treatment assignment, and binary conversion labels. A key characteristic of Criteo is its severely imbalanced treatment assignment (85% treated, 15% control), which is common in deployed ad systems but absent from standard synthetic benchmarks. Since ground-truth CATE is unavailable, we report Qini and policy value at 30% targeting threshold.

Table 1 reports results. Two findings stand out. First, `R-Learner` with tree-based base learners (RF, XGBoost) achieves *negative* Qini coefficients (mean -0.017 and -0.225 respectively), performing worse than random targeting in 4 of 6 runs. This failure does not appear under our synthetic DGPs, which use balanced randomization ($\alpha = 0$). Under high treatment imbalance, the R-Learner's cross-fitted nuisance models for the control arm are estimated on only 15% of training data; tree-based learners overfit to this small control sample and produce unstable residuals. Linear base learners (`R-Learner` [LR]) remain stable (Qini $= 0.74$), confirming that the failure is a learner–imbalance interaction, not a fundamental flaw in the R-Learner objective.

Second, Criteo rankings diverge substantially from synthetic benchmark rankings (Spearman $r = -0.25$ between Criteo Qini and Linear DGP Qini ranks). Methods that perform well under synthetic randomized conditions do not reliably transfer to real deployment settings with treatment imbalance. This is itself

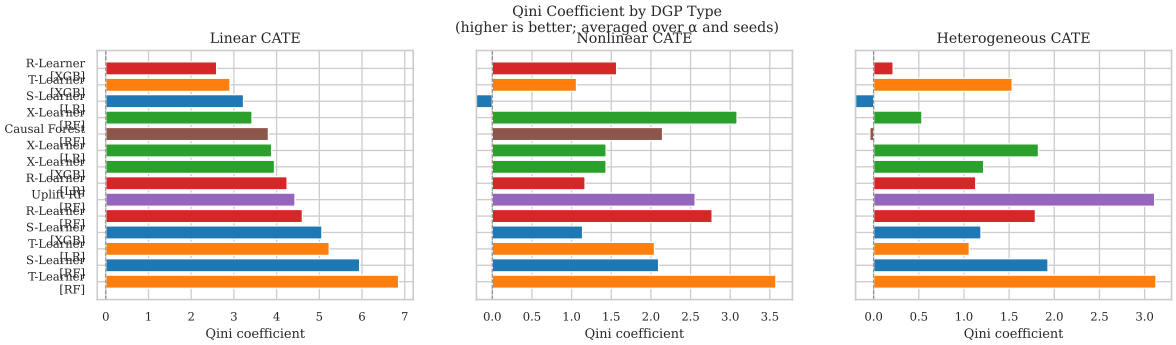

Figure 5: Mean Qini coefficient by DGP type and model. `CausalForest` [RF] achieves competitive Qini but does not translate to superior iROAS efficiency (Figure 7).

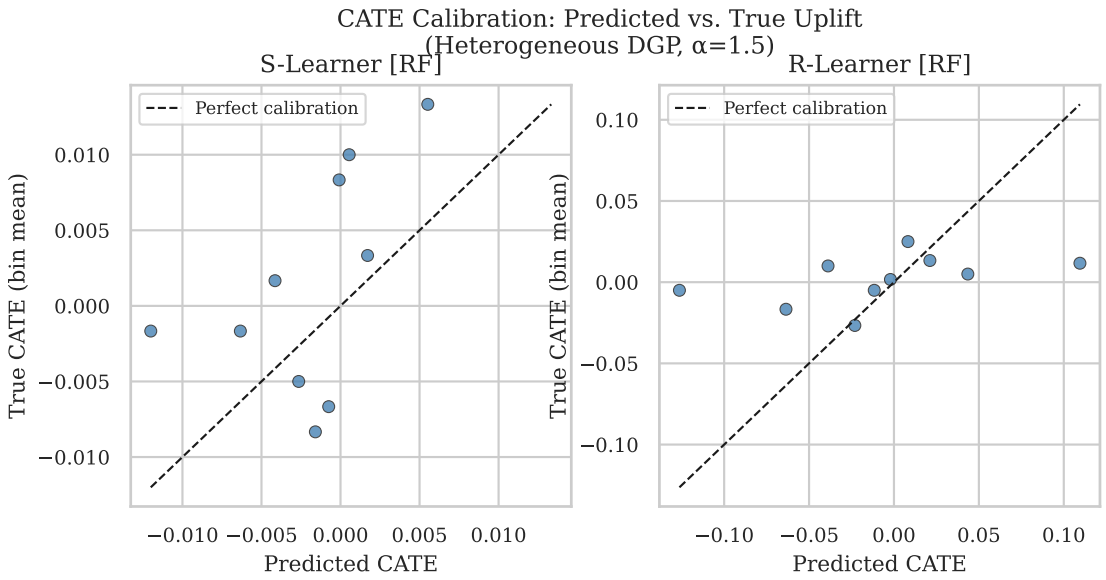

Figure 6: CATE calibration: predicted vs. true mean uplift per decile. `CausalForest` achieves near-diagonal calibration at low α; all methods show calibration breakdown under strong confounding.

an important negative finding: synthetic benchmarks with balanced treatment assignment may mislead practitioners when selecting estimators for real imbalanced settings.

4.5 Summary Table

Table 2 reports all metrics for all methods at the mid-confounding scenario $\alpha = 1.0$, $\rho = 0.05$, averaged over 5 seeds.

5 Discussion

The community is optimizing the wrong objective. Our central finding—that all methods, including the state-of-the-art `CausalForest`, waste 86–89% of ad budget on sure-things—is not primarily a failure of estimation. It is a failure of objective. PEHE measures how accurately a model estimates the treatment effect for each user. But what practitioners actually need is a model that *ranks users by budget-efficiency*: identify the persuadable 40% of users while avoiding sure-things and sleeping dogs. These are related but

Table 1: Real-data results on Criteo Uplift Dataset (500k users, mean ± std over 3 seeds). No ground-truth CATE available; ranking metrics only.

Model	Learner	Qini ↑	AUUC ↑	Policy Value ↑
Causal Forest	RF	0.8177 ± 0.6977	0.8177	0.00225
R-Learner	LR	0.7397 ± 0.0084	0.7397	0.00257
S-Learner	LR	0.6911 ± 0.0000	0.6911	0.00257
T-Learner	XGB	0.6693 ± 0.0000	0.6693	0.00258
X-Learner	LR	0.6369 ± 0.0000	0.6369	0.00249
X-Learner	XGB	0.6289 ± 0.0000	0.6289	0.00244
T-Learner	RF	0.6089 ± 0.0381	0.6089	0.00240
T-Learner	LR	0.5877 ± 0.0000	0.5877	0.00235
Uplift RF	RF	0.5789 ± 0.1867	0.5789	0.00199
X-Learner	RF	0.5526 ± 0.0835	0.5526	0.00233
S-Learner	RF	0.3756 ± 0.0867	0.3756	0.00167
S-Learner	XGB	0.2722 ± 0.0000	0.2722	0.00165
R-Learner	RF	-0.0168 ± 0.2619	-0.0168	0.00074
R-Learner	XGB	-0.2250 ± 0.3600	-0.2250	0.00003

distinct objectives. A model can achieve low PEHE on the bulk of users (who have near-zero CATE) while completely failing to separate the persuadable tail—which is the only group that matters for iROAS. We argue that future work should optimize directly for budget efficiency, not CATE MSE.

Why Qini fails as a sole evaluation criterion. Qini measures whether the model *ranks* users with positive CATE higher than users with zero or negative CATE. Under the Heterogeneous DGP, sure-things (zero CATE) are a large subgroup with high predicted conversion probability. Models that learn to predict conversion—a partially correct proxy for CATE— achieve reasonable Qini by ranking them above lost causes, while still spending substantial budget on them. Wasted spend fraction captures this failure mode; Qini does not. We recommend that any uplift evaluation report wasted spend fraction or iROAS efficiency alongside Qini/AUUC, and provide our benchmark framework to facilitate this.

Implications for `CausalForest`. The inclusion of `CausalForest` as a baseline is important: it is the strongest available estimator by PEHE and Qini, but does not materially outperform simpler meta-learners on wasted spend or iROAS. This is not a criticism of `CausalForest`, which is a correct estimator for CATE—it is evidence that CATE estimation accuracy does not determine budget efficiency under heterogeneous subgroup structure.

Treatment imbalance as a first-class concern. Our Criteo results reveal a failure mode absent from most synthetic benchmarks: under 85% treatment rate, doubly-robust estimators with flexible nuisance models become unstable. `R-Learner` [RF] and `R-Learner` [XGB] achieve negative Qini on real data despite performing well on balanced synthetic data. This suggests that benchmarks should explicitly evaluate models under imbalanced treatment, not only under randomized conditions. At minimum, practitioners should verify performance on a held-out imbalanced split before deploying R-Learner or other cross-fitting approaches.

Practical recommendation. For practitioners: (1) do not evaluate uplift models on Qini alone—report wasted spend fraction and iROAS at your target percentile; (2) if your treatment rate deviates from 50%, test doubly-robust estimators on imbalanced held-out splits before deploying; (3) prefer linear base learners in R-Learner-style models under severe treatment imbalance, or use stabilized cross-fitting with explicit propensity clipping.

Limitations. Our DGPs assume unconfoundedness (all confounding captured in X). Real ad systems may have unmeasured confounders (time-of-day, device, auction price). Binary outcomes abstract away continuous response and multi-touch attribution. The subgroup structure of the Heterogeneous DGP is

stylized; real populations may have more complex CATE landscapes. We release the benchmark framework to facilitate extensions along these axes.

6 Conclusion

We introduced a controlled synthetic benchmark for ad incrementality estimation with three DGPs, six estimator families, and five confounding levels, validated on the Criteo Uplift Dataset. Three findings are central. First, *PEHE systematically mispredicts budget efficiency*: all methods—including `CausalForest`—waste 86–89% of ad budget on sure-thing converters under heterogeneous CATE structure, with less than 3 percentage points separating best from worst. Second, *synthetic rankings do not reliably predict real-data rankings* (Spearman $r = -0.25$ between our benchmark and Criteo), indicating that benchmark design choices—particularly treatment balance assumptions—substantially affect apparent method ordering. Third, *treatment imbalance breaks doubly-robust estimators*: `R-Learner` with tree-based learners achieves negative Qini on Criteo (85% treatment rate), a failure mode invisible under balanced synthetic evaluation.

Together, these findings argue for three changes to how the community evaluates uplift models: (1) replace or supplement Qini with wasted spend and iROAS; (2) include imbalanced treatment conditions in synthetic benchmarks; and (3) treat synthetic and real evaluation as complementary, not interchangeable. We release the full benchmark to support this shift.

The benchmark, all DGPs, models, and experiments are publicly available at [repository URL].

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A DGP Parameter Details

Table 3 summarizes the parameter ranges and default values for all three DGPs.

Table 3: DGP parameter ranges used in the benchmark.

Parameter	Symbol	Range	Default
Confounding strength	α	$\{0.0, 0.5, 1.0, 1.5, 2.0\}$	1.0
Base conversion rate	ρ	$\{0.02, 0.05, 0.10, 0.20\}$	0.05
Number of features	d	—	10
Sample size	n	—	20,000
Random seeds	—	$\{0, 1, 2, 3, 4\}$	42

B Subgroup Structure in Heterogeneous DGP

Table 4: Subgroup definitions in the Heterogeneous DGP.

Subgroup	Weight	True CATE	Baseline CVR	Ad Propensity Bias
Persuadable	40%	$+\eta$	ρ	$+0.3\alpha$
Sure thing	20%	≈ 0	4ρ	0
Lost cause	20%	≈ 0	0.2ρ	0
Sleeping dog	20%	-0.6η	ρ	$+0.4\alpha$

Note that the ad system over-targets both persuadables and sleeping dogs (both have high propensity bias under confounding), reflecting the realistic scenario where the ad delivery algorithm mistakes sleeping dog features for persuadable features when optimizing for conversion probability.

C Additional Results

Full results tables for all $\alpha \times \rho$ combinations are available in the supplementary CSV file `results/uplift_benchmark_v1/resul`

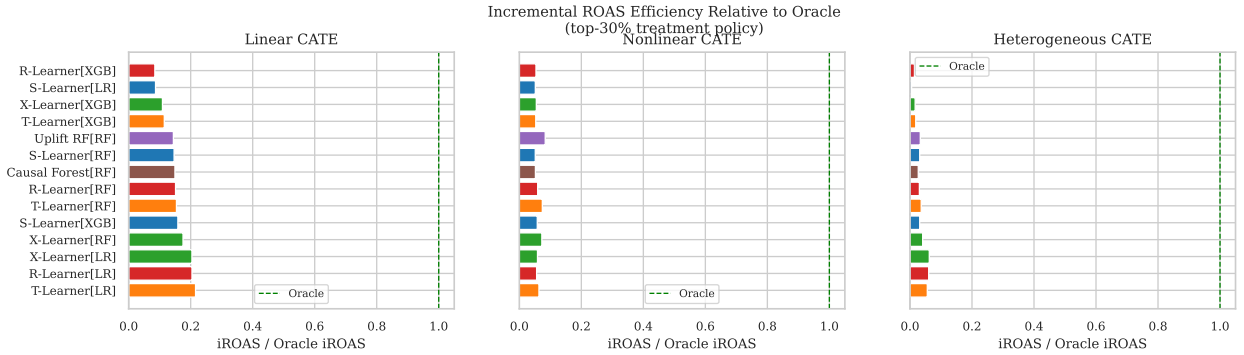

Figure 7: iROAS efficiency (model iROAS / oracle iROAS) under top-30% treatment policy. Values approach 1.0 for a perfect model; values near 0 indicate no incremental value from the targeting policy.

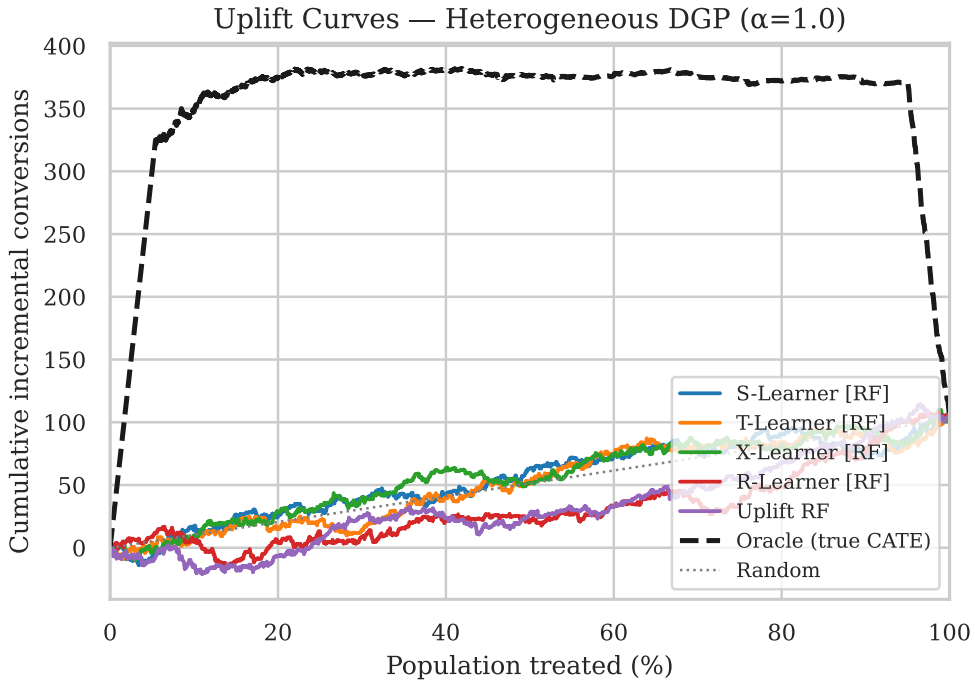

Figure 8: Uplift curves for the Heterogeneous DGP at $\alpha = 1.0$. The oracle (dashed) represents perfect CATE ranking. All meta-learners achieve positive lift over the random baseline, with the R-Learner[RF] closest to oracle.

Table 2: Benchmark results across DGP types (mean ± std over 5 seeds, $\alpha = 1.0$, $\rho = 0.05$)

DGP	Model	Learner	PEHE ↓	Qini ↑	Wasted Spend ↓
Linear CATE					
	Causal Forest	RF	0.408 ± 0.054	5.335	0.871
	R-Learner	LR	0.407 ± 0.053	5.754	0.867
	R-Learner	RF	0.411 ± 0.054	7.552	0.873
	R-Learner	XGB	0.455 ± 0.059	5.188	0.890
	S-Learner	LR	0.409 ± 0.054	7.490	0.850
	S-Learner	RF	0.408 ± 0.054	11.655	0.875
	S-Learner	XGB	0.408 ± 0.054	9.743	0.861
	T-Learner	LR	0.405 ± 0.053	7.703	0.849
	T-Learner	RF	0.408 ± 0.054	13.103	0.875
	T-Learner	XGB	0.429 ± 0.060	5.370	0.873
	Uplift RF	RF	0.415 ± 0.056	7.064	0.879
	X-Learner	LR	0.407 ± 0.053	4.972	0.865
	X-Learner	RF	0.407 ± 0.054	3.667	0.866
	X-Learner	XGB	0.423 ± 0.056	7.758	0.882
Nonlinear CATE					
	Causal Forest	RF	0.410 ± 0.055	0.828	0.889
	R-Learner	LR	0.410 ± 0.055	0.501	0.880
	R-Learner	RF	0.414 ± 0.055	1.026	0.888
	R-Learner	XGB	0.458 ± 0.061	-0.012	0.897
	S-Learner	LR	0.409 ± 0.055	0.103	0.834
	S-Learner	RF	0.410 ± 0.055	1.285	0.888
	S-Learner	XGB	0.410 ± 0.055	0.198	0.879
	T-Learner	LR	0.410 ± 0.055	1.014	0.870
	T-Learner	RF	0.411 ± 0.055	2.377	0.883
	T-Learner	XGB	0.432 ± 0.061	0.570	0.880
	Uplift RF	RF	0.421 ± 0.060	2.267	0.882
	X-Learner	LR	0.410 ± 0.055	0.851	0.882
	X-Learner	RF	0.411 ± 0.055	1.573	0.883
	X-Learner	XGB	0.424 ± 0.056	0.823	0.893
Heterogeneous CATE					
	Causal Forest	RF	0.466 ± 0.099	0.131	0.875
	R-Learner	LR	0.465 ± 0.099	0.241	0.873
	R-Learner	RF	0.470 ± 0.100	0.241	0.871
	R-Learner	XGB	0.518 ± 0.109	0.038	0.878
	S-Learner	LR	0.465 ± 0.099	-0.338	0.887
	S-Learner	RF	0.465 ± 0.099	-0.025	0.874
	S-Learner	XGB	0.466 ± 0.099	-0.072	0.872
	T-Learner	LR	0.465 ± 0.099	0.085	0.872
	T-Learner	RF	0.467 ± 0.099	0.186	0.872
	T-Learner	XGB	0.492 ± 0.110	-0.109	0.870
	Uplift RF	RF	0.472 ± 0.101	0.143	0.868
	X-Learner	LR	0.465 ± 0.099	0.146	0.881
	X-Learner	RF	0.466 ± 0.099	-0.015	0.870
	X-Learner	XGB	0.484 ± 0.104	-0.011	0.881